12. $\frac{\left(x+\frac{1}{2}\right)}{2}\sqrt{1+x+x^2}+\frac{3}{8}\log\left|\left(x+\frac{1}{2}\right)+\sqrt{1+x+x^2}\right|+c$

13. $\frac{x}{2}\sqrt{x^2-2}-\log\left|x+\sqrt{x^2-2}\right|+c$ 14. $\frac{1}{4}\left[2x\sqrt{4x^2-5}-5\log\left|2x+\sqrt{4x^2-5}\right|\right]+c$

15. $\left(\frac{x+1}{2}\right)\sqrt{2x^2+4x+1}-\frac{\sqrt{2}}{4}\log\left|\sqrt{2}(x+1)+\sqrt{2x^2+4x+1}\right|+c$

16. $\frac{x^2}{2}-\frac{x}{2}\sqrt{x^2-1}+\frac{1}{2}\log\left|x+\sqrt{x^2-1}\right|+c$

Exercise 2.8

I:1. $\frac{1}{2}\left[e^2-1\right]$ 2. $\frac{1}{6}$ 3. $\frac{1}{2}\log\left[\frac{5}{2}\right]$ 4. $\log\left[\frac{1+e^3}{2}\right]$

5. $\frac{1}{2}\left[e-1\right]$ 6. $\frac{3}{8}$ 7. $\log\left[\frac{11}{5}\right]$ 8. 2 9. $\frac{1}{2}\left[2\log 2-1\right]$

II:1. 37 2. 4 3. 1 4. $c=4$

Exercise 2.9

1. 0 2. $\frac{\pi}{2}$ 3. 0 4. $\frac{\pi}{4}$ 5. 0 6. $\frac{16}{5}$

Exercise 2.10

1.(i) 6 (ii) $\frac{105\sqrt{\pi}}{16}$ (iii) $\frac{6!}{m^7}$ (iv) $\frac{3}{128}$ (v) $\left(2^6\right)5!$ 2. $\frac{1}{4}$

Exercise 2.11

1. $\frac{9}{2}$ 2. 4 3. 14 4. $\frac{1}{3}$

Exercise 2.12

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
(b)	(c)	(a)	(a)	(a)	(b)	(b)	(a)	(d)	(c)	(b)	(b)	(b)	(b)	(c)
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
(d)	(c)	(c)	(b)	(a)	(b)	(b)	(c)	(a)	(a)	(a)	(b)	(d)	(b)	(c)

Miscellaneous problems

1. $\frac{2}{15}\left[(x+2)^{\frac{3}{2}}+(x+3)^{\frac{3}{2}}\right]+c$ 2. $\frac{1}{5}\log\left|\frac{2+x}{1-2x}\right|+c$ 3. $\frac{1}{4}\log\left|\frac{e^x+1}{e^x+5}\right|+c$

4. $\frac{x}{2}\sqrt{2x^2-3}-\frac{3\sqrt{2}}{4}\log\left|\sqrt{2}x+\sqrt{2x^2-3}\right|+c$

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